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Towards a two-stage low-cost soil carbon estimation over large area using proximal sensing and digital soil mapping

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ABSTRACT

Accurate and scalable estimation of soil organic carbon (SOC) is essential for sustainable land management and climate mitigation, yet conventional laboratory-based methods remain costly and impractical for large-area monitoring. This study presents a novel two-stage framework that integrates low-cost proximal red, green, and blue (RGB) soil imaging, deep learning, and machine learning-based digital soil mapping (DSM) to enable regional SOC prediction at 100 m resolution. In the first stage, SOC was estimated from RGB images of 405 surface soil samples collected across multiple agro-climatic zones of West Bengal, India, using a random forest model based on color features and a fine-tuned Visual Geometry Group 16-layer (VGG16) convolutional neural network. The deep learning model achieved superior performance, with a 37.5 % higher validation R^2 and 13.1 % lower mean absolute error (MAE) compared to the color feature-based model. In the second stage, RGB-derived SOC predictions were integrated with multi-source environmental covariates within a DSM framework, yielding regional SOC maps that slightly outperformed a benchmark wet chemistry-based DSM with 8.8 % higher test R^2 and 32.1 % lower MAE. The maps revealed strong agro-climatic control on SOC distribution, with higher SOC in the northern Terai Zone (1.1–2.0 %) and lower SOC in the Red and Laterite Zone (<0.8 %), although both models showed limited sensitivity for low SOC (<0.5 %) soils. A benefit–cost analysis based on the Analytic Hierarchy Process demonstrated that the proposed RGB–DSM framework is more than twice as efficient as conventional wet chemistry-based mapping in terms of cost, time, and environmental safety. The study establishes a scalable and operationally viable pathway for SOC monitoring using consumer-grade imaging and deep learning, with strong potential for deployment in resource-limited regions and digital measurement, reporting, and verification systems.

1. Introduction

Soil organic carbon (SOC) is a crucial soil health and agricultural sustainability determinant that affects many biophysical processes such as nutrient cycling, water retention, soil aggregation, and microbial activity (Weil and Magdoff, 2004). In addition to agronomic importance, SOC is one of the largest stocks in the global carbon cycle and makes an important contribution to climate regulation (Luo et al., 2019; Beillouin et al., 2023). Its management is closely linked to the United Nations Sustainable Development Goals (SDGs), particularly SDG 2 (Zero Hunger) to support target 2.4 (sustainable food production), SDG 13 (Climate Action) for carbon sequestration relevant to target 13.2 on integrating climate change measures, and SDG 15 (Life on Land), specifically target 15.3 on land restoration and achieving land degradation neutrality (United Nations, 2015). Thus, accurately measuring and predicting the spatial distribution of SOC is essential for soil resource management and climate change mitigation (Vågen and Winowiecki, 2013). Agricultural lands, especially are inherently low in SOC content (Lal et al., 2015) and have lost around 30–50 % of carbon due to intensive cultivation (Blanco-Canqui, 2025). The global cropland soils

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have around 29–65 Pg of carbon storage potentials (Padarian et al., 2022), and may play an important role in the voluntary carbon market (Newell-Price et al., 2022). Inclusion of cost-effective as well as reliable alternate solutions in the Digital Measurement, Report and Verification (DMRV) process (Santier, 2023) will aid in assessing the gap existing in land carbon levels and better manage the declining soil health through adaptation of carbon sequestering management practices (Schreiner-McGraw et al., 2024).

Traditional laboratory-based methods for SOC estimation, such as wet oxidation and dry combustion, are widely regarded as reliable but present significant limitations (Nelson and Sommers, 1996). These methods are often time-consuming, expensive, and require specialist infrastructure and trained personnel, which limit their use for SOC mapping at scale and resolution, especially in resource-limited settings (Aitkenhead et al., 2015; Salo et al., 2016). As a result, there is an urgent need for fast, scalable and cost-effective estimation methods that can estimate SOC on large spatial scales, which have prompted research into the best non-destructive approaches for estimating SOC over vast spatial extent (Liang et al., 2019).

Proximal sensing technologies and advanced data analytics, specifically those that use machine learning (ML) and deep learning (DL) algorithms, have shown potential for SOC estimation (Zhao et al., 2022; Bai et al., 2022). Soil color, a key diagnostic property in soil classification, plays a crucial role in estimating SOC because SOC content strongly influences soil darkness or lightness. In the visible spectrum (400–700 nm), soil color reflects the combined effects of SOC, Fe- oxides/hydroxides, and light-colored minerals like quartz, feldspars, or carbonates and also contributed by texture and other constituents (Konen et al., 2003; Viscarra Rossel et al., 2006; Swetha and Chakraborty, 2021; Brown and O'Neal, 1923). While the traditional Munsell color chart remains widely used for determining soil color, its accuracy is limited by operator subjectivity, lighting conditions, and the discrete nature of color chips, unlike the continuous variation captured through reflectance measurements (Sánchez-Marañón et al., 2015; Swetha et al., 2022). Alternatively, red, green and blue (RGB) image-based photography using digital cameras on smartphone platforms can also provide a low-cost and practical solution for SOC and other soil property evaluations (Levin et al., 2005; Swetha et al., 2020; Gorthi et al., 2021; Taneja et al., 2021, 2022). Notably, RGB images, when processed appropriately, capture the surface color and texture characteristics that are associated with soil organic matter content (Viscarra Rossel et al., 2008; de Oliveira Morais et al., 2019; Gholizadeh et al., 2020; Heil et al., 2020, 2022; Dasgupta et al., 2024). Fu et al. (2020) demonstrated that cell phone captured soil RGB images, when corrected for illumination and processed across color spaces, effectively predicted soil organic matter, with soil moisture $>$ 10 % significantly reducing accuracy unless explicitly included in the prediction model. While many studies have been conducted using hyperspectral and multispectral sensors (Hbirkou et al., 2012; Hobley et al., 2018), studies using RGB imagery, especially on low-cost consumer-grade mobile devices, are few in the context of SOC prediction.

In parallel, digital soil mapping (DSM) methodologies have been increasingly adopted to predict spatial distributions of soil properties by integrating observed data with spatially explicit environmental covariates (McBratney et al., 2003; Malone et al., 2016; Minasny and McBratney, 2016). DSM is closely associated with soil mapping and modelling, utilizing georeferenced soil information derived from spatial soil databases. Based on the SCORPAN framework, DSM approaches incorporate factors such as climate, vegetation, topography, and parent material, and have traditionally relied on point data obtained from laboratory analysis (Dasgupta et al., 2023; Das et al., 2025).

Over the past decade, DSM has undergone a major methodological transition driven by advances in ML, DL, and cloud-based geospatial analytics. Contemporary DSM frameworks increasingly rely on data-driven models capable of capturing nonlinear relationships between soil properties and environmental covariates across multiple spatial scales. Random Forest (RF), Gradient Boosting, Extreme Gradient Boosting (XGBoost), Support Vector Regression (SVR), and deep neural networks have become dominant modelling paradigms due to their superior predictive accuracy and robustness compared to traditional geostatistical approaches (Padarian et al., 2019; Wadoux et al., 2020; Hengl et al., 2023). Ensemble learning strategies that combine multiple algorithms are now widely adopted to improve generalisation and uncertainty quantification in spatial soil prediction (Taghizadeh-Mehrjardi et al., 2021; Zhang et al., 2022).

Recent studies have demonstrated the effectiveness of hybrid ML–geostatistical frameworks, such as regression-kriging, RF-kriging, and deep residual kriging, which integrate spatial autocorrelation with nonlinear covariate learning (Tunçay et al., 2023). These models use the strengths of both deterministic prediction and spatial interpolation, enabling improved mapping of SOC and other soil properties at fine spatial resolution. In parallel, DL models such as convolutional neural networks (CNNs), long short-term memory networks (LSTMs), and graph neural networks (GNNs) are increasingly being explored for spatial soil inference, particularly when large volumes of remote sensing data are available (Padarian et al., 2019; Zeng et al., 2022; Bao et al., 2024). The rapid growth of global soil databases and Earth observation platforms has further accelerated the adoption of large-scale ML-based soil mapping. Initiatives such as SoilGrids (Poggio et al., 2021), OpenLandMap (Hengl et al., 2025), and GlobalSoilMap have demonstrated the feasibility of producing continental to global soil property maps using ensemble ML frameworks trained on millions of harmonized soil observations. At the same time, proximal sensing and low-cost sensing technologies are increasingly being integrated into DSM workflows to overcome limitations of laboratory-based soil analysis. The potential integration of proximal sensing-based predictions into DSM workflows offers a new pathway for enhancing scalability while reducing reliance on laboratory inputs (Viscarra Rossel et al., 2010; Silva et al., 2016; Lavanya et al., 2024). Portable spectroscopy, RGB imaging, smartphone sensors, and in-field spectrometers are now being used to generate dense point-scale soil information that can be spatially upscaled using ML-based DSM (Adler et al., 2022; Sun et al., 2023). These developments have enabled a new generation of scalable, cost-effective, and near-real-time soil monitoring systems that align with the requirements of precision agriculture, carbon accounting, and digital measurement, reporting, and verification (DMRV) frameworks. The present study introduces a two-stage modelling framework for SOC prediction and spatial mapping. The first stage involves SOC estimation at the point scale at centimeter spatial resolution using RGB soil images, processed through two-step modelling approaches: (i) a random forest (RF) model (Breiman, 2001) based on average image-derived color features, and (ii) a fine-tuned CNN (Li et al., 2021) using the Visual Geometry Group 16-layer (VGG16) architecture (Simonyan and Zisserman, 2014). The second stage involves DSM, wherein the SOC predictions from the RGB-based VGG16 model were used as target variables, and a suite of geospatial environmental predictors, derived from satellite and terrain data, was used to develop spatial SOC distribution maps. The objectives of this work were to: (1) evaluate the effectiveness and accuracy of RGB soil imaging combined with ML and DL techniques for predicting SOC, and (2) develop and compare digital SOC maps generated using both RGB-derived and laboratory wet chemistry-based SOC estimates, (3) landscape-scale assessment of DSM-based SOC with respect to in situ SOC from proximal sensing. The working hypothesis of this research was that SOC predictions based on RGB soil imaging in combination with DL models such as VGG16 can remove the reliance on laboratory wet chemistry-based SOC estimates in order to have extensive and quicker but reliable in situ SOC estimates over a large number of locations. It is further hypothesized that spatial SOC maps created from (soil RGB-image + DL model)-based predictions would accurately depict the SOC spatial variability, providing further support for the scalability of proximal photo-analytics for effective soil quality monitoring and DSM across various landscape contexts.

2. Materials and methods

2.1. Study area

Soil samples were collected from various agro-climatic zones across the West Bengal state, located in eastern India between latitudes 21.68°N and 24.80°N and longitudes 86.11°E and 89.42°E (Dasgupta et al., 2022, 2023, 2024) (Fig. 1). The southern region of the state comprises three agro-climatic zones: the Coastal and Saline Zone (CSZ) in the southeast (Sarkar et al., 2022), the Old Alluvial Zone (OAZ) in the south-central area, and the Red and Laterite Zone (RLZ) in the south-west. Soils in the CSZ are saline, nutrient-poor, poorly drained, and belong to the great groups such as Haplaquepts, Paleudults, Halaquepts, and Fluvaquents (Nayak et al., 2001). In contrast, RLZ soils are highly weathered, leached, acidic, and deficient in available nitrogen, falling under Tropaquepts, Paleaustalfs, and Haplustalfs. Samples were also collected from the central part of the state, which includes the New Alluvial Zone (NAZ) in the central-east and the OAZ in the central-west. These soils, formed from alluvial deposits of the Ganges River system, are fertile and primarily classified as Eutrochrepts, with occurrences of Fluvaquents, Haplaquepts, and Paleustalfs. Additionally, samples were collected from the northern Terai Zone (TZ), where soils are derived from lesser Himalayan schists and phyllites. These soils are well-drained, rich in organic matter, acidic in nature, and categorized under Haplaquepts and Ustifluvents.

Climatic conditions across the study area vary widely: the CSZ experiences a tropical monsoon climate (Am), the RLZ a tropical wet and dry climate (Aw), the central zones (NAZ and OAZ) a humid subtropical climate (Cwa), and the TZ a subtropical highland climate (Cwb) (Peel et al., 2007). Annual rainfall ranges from approximately 100–125 cm in the southwest to 200–400 cm in the northern regions. Topographically, the terrain shifts from steep slopes (~32 %) in the hilly north to flat Gangetic alluvial plains with slopes of < 1 % in the south. Rice is the predominant crop across the state, cultivated in both winter and summer seasons (Mondal et al., 2020). Other major crops include jute, oilseeds, potatoes, vegetables, and fruits such as mango, banana, and jackfruit. In the northern region, high-quality tea is cultivated along with crops such as oranges, pineapples, ginger, and cardamom (Dasgupta et al., 2022).

2.2. Sample collection and chemical analysis

A total of 405 surface soil samples (0–20 cm) were collected from fallow fields following crop harvest during November–December of 2019–2020 using a handheld trowel, following standard procedures outlined by the Soil Survey Staff (2024) (Fig. 1b). A stratified random sampling approach was employed based on variations in soil types (Entisols, Alfisols, Inceptisols), parent materials (recent, old, or deltaic alluvial deposits, schist-phyllites, and granite-gneiss), and soil productivity potential. A minimum distance of around 6 km was maintained between any two sampling locations, ensuring sufficient intra-zonal variations between collected samples. A total of 195 samples were collected from the CSZ and RLZ, 147 samples from the NAZ and OAZ, and 63 samples from the TZ. At each sampling point, four subsamples were collected, thoroughly mixed, and transported in zip-lock plastic bags. The geographic coordinates of the sampling sites were recorded using a Garmin E-Trex GPS device (Garmin, Olathe, KS, USA), which has a positional accuracy of 1 m. In the laboratory, the soil samples were air-dried, ground, and passed through a 2 mm sieve before being stored for subsequent chemical analysis. SOC content (%) was determined using the chromic acid oxidation method as described by Nelson and Sommers (1996).

2.3. Proximal soil image acquisition

Around 25 g of soil was evenly spread in a petri dish to a uniform thickness of 10 mm across the surface. To avoid microtopographic shading effects, the soil surface was gently levelled with a rounded bottle stopper. The petri dishes containing evenly spread soil samples were placed in a custom-made imaging chamber measuring 12 cm × 8 cm × 5 cm (Fig. 2). White light-emitting diode (LED) strips, powered by a DC power adaptor illuminated the chamber at an intensity of 1500 lux. A white reference card was also placed near the soil samples

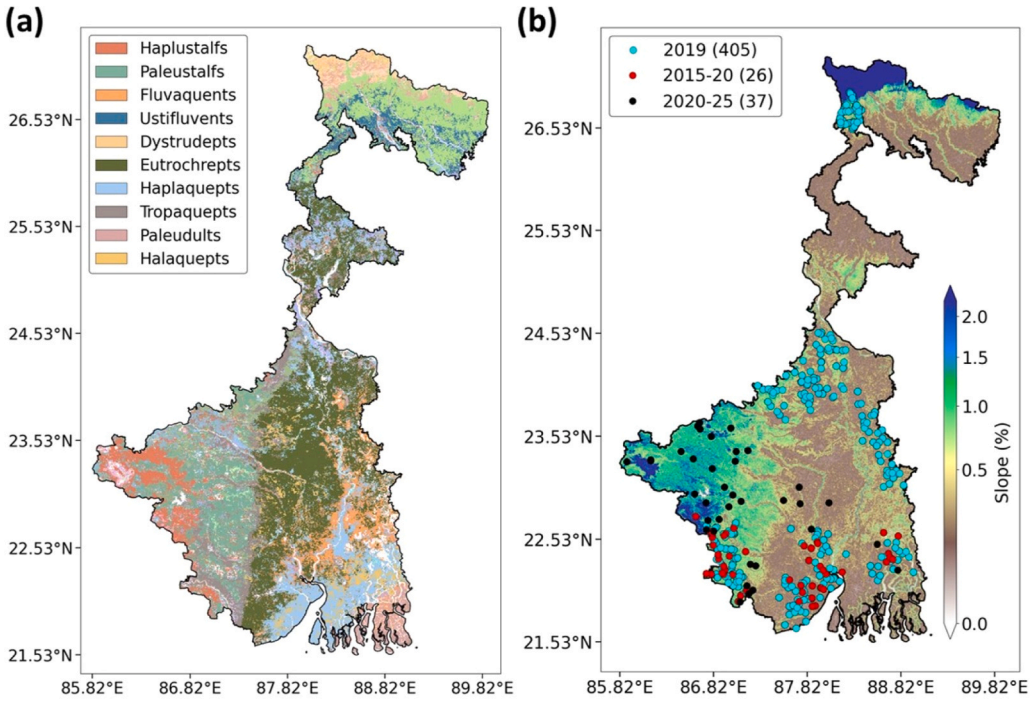


Fig. 1. (a) Location of the study area in West Bengal, India, showing OpenLandMap USDA soil great groups [accessed from Google Earth Engine, Hengl and Nauman (2018)], (b) Distribution of sampled locations collected during 2015–2020, 2019 and 2020–2025. The number of samples collected during each timeframe is shown in parentheses. The slope (%) variation within the study area is also shown in a color gradient.

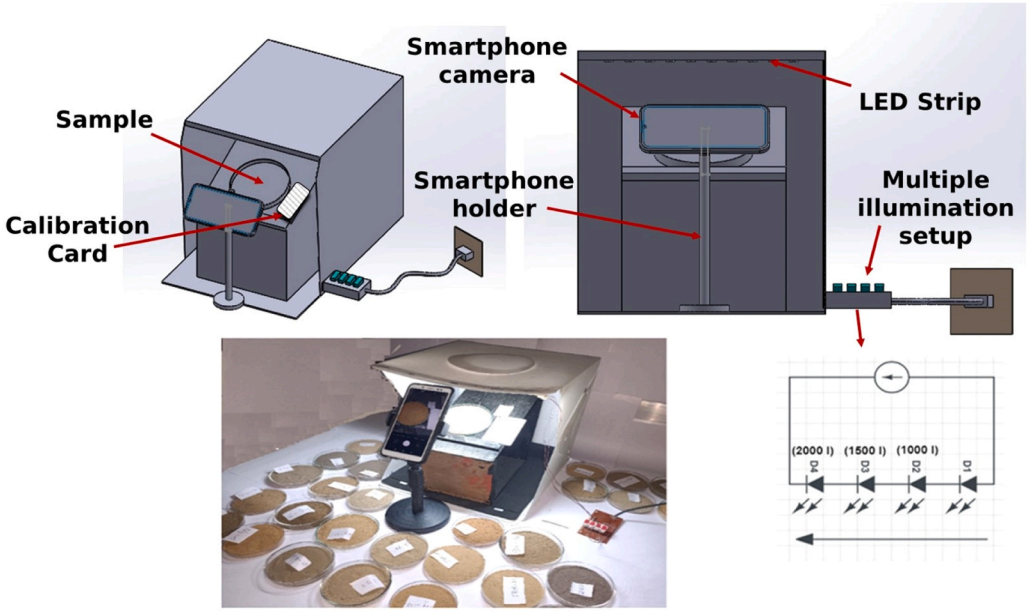


Fig. 2. Image acquisition setup used for capturing red, green, and blue (RGB) images of soil samples collected from different agro-climatic zones of West Bengal.

for RGB image calibration. As capturing images from different imaging devices may provide different color information for the same sample (Yang et al., 2021), we utilized a single smartphone based imaging device for the current study. A Redmi Note 6 Pro (Xiaomi, Beijing, China) mobile phone with 6 GB of random access memory (RAM) and 64 GB storage was mounted on a holder to capture images at a 45° acute angle. To ensure consistent Field of View (FOV), images were captured from a fixed distance of 10 cm above the soil surface using fixed a smartphone camera holder. The mobile phone features a rear 12 megapixel primary camera with f /1.9 aperture size, 1/2.55 in. sensor size, and 1.4 μm pixel size, as well as a 5 megapixel depth camera on the rear secondary camera with f/2.2 aperture size. RGB photographs of soil samples were captured in manual mode with an ISO of 100, shutter speed of 1/100 s, aperture size of f/1.9 (primary camera). The white balance was fixed to daylight mode and all automatic corrections (exposure, contrast, sharpness, brightness) and flash were disabled. For each sample, two image replicates were captured by rotating the petri dish by 90°. All captured images were saved to the memory card of the mobile phone in Joint Photographic Experts Group (JPEG) format.

2.4. Color space features extraction

All captured RGB images were resized to 1280 × 960 pixels in OpenCV python package, employing area-based interpolation to preserve pixel relationships. The images were then cropped to a size of 224 × 224 pixels from the middle portion of the image. Average R, G, and B channel values in the form of digital numbers were computed for each image. Various tristimulus values represented by the relative amount of three primary colors (red, green, and blue), to visually match a given color (Prasad et al., 1996). Different color models, including RGB, HSV, CIE XYZ (Ripamonti et al., 2004), CIE Lab (CIE, 2004), and CIE LUV (CIE, 2004) were used. More details about these color space models are provided in Viscarra Rossel et al. (2006) and Gholizadeh et al. (2020). All the color space models were implemented using the cv2 package in Python. Since there is a high correlation between R, G, and B values, three statistically independent components related to the hue, intensity and chromatic information, termed as H_{RGB}, I_{RGB}, and S_{RGB}, respectively (Viscarra Rossel et al., 2006), were also extracted from each image. Furthermore, 16 additional color space indices were computed (Table 1).

Table 1
Details of the color features used for predicting soil organic carbon of samples collected from West Bengal, India.

Feature	Formula	Reference
Normalized red	R/(R+G+B)	Xu et al. (2010)
Normalized green	G/(R+G+B)	Xu et al. (2010)
Normalized blue	B/(R+G+B)	Xu et al. (2010)
Normalized green-red difference index	(G-R)/(G+R)	Tucker (1979)
Dark green color index	(H − 60)/60 + (1 − S) + (1 − V)/3	Karcher and Richardson (2003)
Kawashima index	(R-B)/(R+B)	Kawashima and Nakatani, (1998)
Red green ratio index	R/G	Xu et al. (2010)
Difference between green and blue	G-B	Xu et al. (2010)
Difference between green and red	G-R	Xu et al. (2010)
Difference between normalized green and normalized blue	G-B	Xu et al. (2010)
Color feature index	G/B	Xu et al. (2010)
Redness index	$R^2/(B \times G^3)$	Barron and Torrent (1984)
Colouration index	(R-G)/(R+G)	Mathieu et al. (1998)
Saturation index	(R-B)/(R+B)	Mathieu et al. (1998)
Brightness index	sqrt {($B^2 + G^2 + R^2$)/3}	Mathieu et al. (1998)
Hue index	(2 x R – G -B)/(G-B)	Mathieu et al. (1998)

2.5. Proximal image-based SOC predictions

Of the 405 collected samples, initially 305 were used for image-based SOC prediction using ML and DL models, while the remaining 100 were reserved for testing the DSM model (hereafter referred to as the test dataset). The 305 samples were randomly split into a training set (~70 %, n = 214) and a validation set (~30 %, n = 91). To maintain consistent illumination condition across the samples, a global RGB histogram equalization and min-max normalization was applied on each image.

For the ML approach, average color features from each image were used to train an RF model (Breiman, 2001) using the *PyCaret* package in Python. To address multi-collinearity, variables with a Pearson's correlation coefficient > 0.3 were removed during RF training. RF model training was conducted using a 5-fold cross-validation framework on the

training set (n = 214). Additionally, an external validation was also performed using the validation set (n = 91). Various hyperparameters of the RF model (such as number of trees, maximum features, maximum leaf nodes, minimum samples per leaf, minimum samples per split) were automatically optimized utilizing random grid search from the default grid of hyperparameters defined in the *PyCaret* package.

For the DL approach, a pre-trained VGG16 convolutional neural network (CNN) (Simonyan and Zisserman, 2014), originally developed for object detection, was adapted for proximal RGB-based SOC prediction using *TensorFlow* package in python. The VGG16 model contains 138 million parameters, comprising 13 convolutional layers followed by three fully connected (FC) layers. The original top layer was removed, and the saved weights of the first 13 convolutional layers (excluding the FC layers) were loaded and frozen. To reduce the spatial dimension into a compact feature vector, a new global average pooling layer was added. Followed by that a fully connected head with six dense layers having 512, 256, 128, 64, 32, and 16 neurons, respectively were added each utilizing a rectified linear unit (ReLU) activation. The final output layer for SOC prediction used a linear activation. The model training was done in two stages. In the first stage, model weights and biases in the new FC layers were optimized using the Adam optimizer with a learning rate of 0.0001, while the weights and biases of base layers were kept fixed. In the second stage, few last layers of the base model along with new FC layers were fine-tuned with smaller learning rate (0.000001) to adapt the model for the new task of SOC prediction. The model was trained on the training set for 100 epochs. To prevent overfitting, early stopping was applied with a patience level of 10 based on mean squared error of the validation set (n = 91).

2.6. Digital soil mapping of SOC

Various environmental covariates related to the SCORPAN soil forming factors (McBratney et al., 2003) were incorporated into the digital soil mapping (DSM) (Table 2). All datasets were processed using the Google Earth Engine cloud-based platform. Most covariates related to soil, climate, and relief utilized in this study were static in time. These included soil physical and chemical properties from the Global SoilGrids dataset (Hengl et al., 2017), soil great groups, soil salinity (Ivushkin et al., 2019), and soil hydrological properties (Simons et al., 2020). Long-term mean temperature and precipitation data from the WorldClim Version 1 climatological datasets were used to represent climatic variation in the study area. Additionally, topographical parameters such as geomorphological forms, first-order rate of change across the elevation gradient, and ruggedness were obtained from the Geomorpho90m dataset (Amatulli et al., 2020).

Apart from the static covariates, time-series dynamic covariates in the form of various vegetation indices related to vegetation growth were derived from Landsat 7 Level 2 surface reflectance data for the period 2000–2018. Proper scale and offset factors were applied to different spectral bands of Landsat data. A normalized difference vegetation index (NDVI)–based threshold approach was used to generate bare-soil (BS) and active-crop (AC) masks dynamically for each scene in the Landsat 7 time-series image collection. For BS images, a NDVI threshold between 0.01 and 0.25 was applied, whereas for AC images, a NDVI threshold > 0.35 was used. Subsequently, from the BS and AC reflectance images, various maximum-value reflectance composite images were generated considering the compositing window of one year. Subsequently, various vegetation indices were also computed on an annual timescale (Table 3). The long-term averages of these indices (over 18 years) were then computed and included as additional covariates in the DSM model for SOC prediction.

The SOC contents estimated from proximal RGB image data (from previous stage) along with different environmental covariates were used to build a DSM framework. For comparing the spatial SOC estimates from the two-stage prediction model, a separate model trained on satellite covariates and actual wet chemistry based SOC content was used as a benchmark. To train the model in second stage, a total of 305 soil samples (considering both calibration and validation datasets, utilized in the previous stage) were used to train the DSM. A RF model with 5-fold cross-validation was utilized for training the DSM model in

Table 2
Geospatial datasets used for building the digital soil map for the soils of West Bengal, India.

Covariate Details	Data	Spatial Resolution
Vegetation indices derived from atmospherically corrected surface reflectance and surface temperature	USGS Landsat 7 Level 2, Collection 2, Tier 1	30 m
Bioclimatic variables such as annual mean temperature (BIO 01), mean diurnal range of monthly minimum and maximum temperature (BIO 02), annual precipitation (BIO 12).	WorldClim version 1 (Hijmans et al., 2005)	1 km
Soil pH, bulk density in Mg/m^3 (BD), cation exchange capacity in cmol $(p^+)/kg$ (CEC), sand, silt, clay	Global SoilGrids (Poggio et al., 2021)	250 m
OpenLandMap USDA taxonomy soil Great Groups	Hengl and Nauman (2018)	250 m
Available water content (Wcavail) in m^3/m^3; and hydrologic soil group	HiHydroSoil v2.0 (Simons et al., 2020)	250 m
Global soil salinity map 2016	Ivushkin et al. (2019)	250 m
Geomorphological forms (geom), slope, aspect, stream power indices (SPI), compound topographic index (CTI), terrain ruggedness index (TRI), roughness, and topographic position index (TPI)	Geomorpho90m (Amatulli et al., 2020)	100 m

Table 3
Different vegetation indices computed over bare soil (BS) and active crop (AC) image composites for the soils of West Bengal, India.

Name of indices*	Formula	Reference
Computed over Active Crop (AC) composite image		
NDWI	$\frac{(\rho NIR - \rho SWIR2)}{(\rho NIR + \rho SWIR2)}$	Chen et al. (2005)
kNDVI	$\tanh(NDVI)^2$	Wang et al. (2023)
SLAVI	$\frac{\rho NIR}{\rho Red + \rho SWIR1}$	Lymburner et al. (2000)
SAVI	$\frac{(\rho NIR - \rho Red)}{(\rho NIR - \rho Red + 0.5)} \cdot 1.5$	Huete (1988)
EVI	$\frac{2.5 \cdot (\rho NIR - \rho Red)}{(\rho NIR + 6 \cdot \rho Red - 7.5 \cdot \rho Blue + 1)}$	Huete et al. (2002)
Computed over Bare Soil (BS) composite image		
BI2	$\frac{\sqrt{(\rho Red \cdot \rho Red) + (\rho Green \cdot \rho Green) + (\rho NIR \cdot \rho NIR)}}{3}$	Escadafal (1989); Gholizadeh et al. (2018)
RI	$\frac{(\rho Red \cdot \rho Red)}{(\rho Green \cdot \rho Green \cdot \rho Green)}$	Pouget et al. (1990)
CRC	$\frac{(\rho SWIR1 - \rho Green)}{(\rho SWIR1 + \rho Green)}$	Sullivan et al. (2006)
NDTI	$\frac{(\rho SWIR1 - \rho SWIR2)}{(\rho SWIR1 + \rho SWIR2)}$	Van Deventer et al. (1997)
SATVI	$\left(\frac{\rho SWIR1 - \rho Red}{\rho SWIR1 + \rho Red + 1} \cdot 2\right) - \frac{\rho SWIR2}{2}$	Marsett et al. (2006)
LSWI	$\frac{(\rho NIR - \rho SWIR1)}{(\rho NIR + \rho SWIR1)}$	Xiao et al., (2005)

* NDWI: Normalized Difference Water Index, kNDVI: kernel Normalized Difference Vegetation Index, SLAVI: Specific Leaf Area Vegetation Index, SAVI: Soil Adjusted Vegetation Index, EVI: Enhanced Vegetation Index, BI2: Brightness Index 2, RI: Redness Index, CRC: Crop Residue Cover, NDTI: Normalized Difference Tillage Index, SATVI: Soil-Adjusted Total Vegetation Index, LSWI: Land Surface Water Index.

PyCaret. To reduce multi-collinearity impact on the DSM model, variables having Pearson's correlation coefficient of > 0.3 were excluded from further analysis. A separate test set ($n = 100$), not utilized in any prior modelling was used for external evaluation of the model.

In addition, five years maximum Value Composite (MVC) of vegetation indices, generated similarly from raw Landsat 7 Level 2 surface reflectance were utilized along with the static covariates to predict the SOC content across two separate time periods of 2015–2020 and 2020–2025 using the trained proximal RGB based DSM model. For the validation of the predicted SOC content during these two timeframes, additional 26 observations in 2015–2020 and 37 observations in 2020–2025 were utilized. These samples were not utilized anywhere in the proximal or DSM model.

2.7. Statistical analysis

To evaluate the performance of RF, VGG16 models fitted for proximal RGB image-based point-scale SOC prediction, as well as the RF-based DSM model for spatial predictions; different accuracy matrices were computed. Coefficient of determination (R^2) (Eq. 1), and mean absolute error (MAE) (Eq. 2) for training, validation, and testing datasets were calculated as follows:

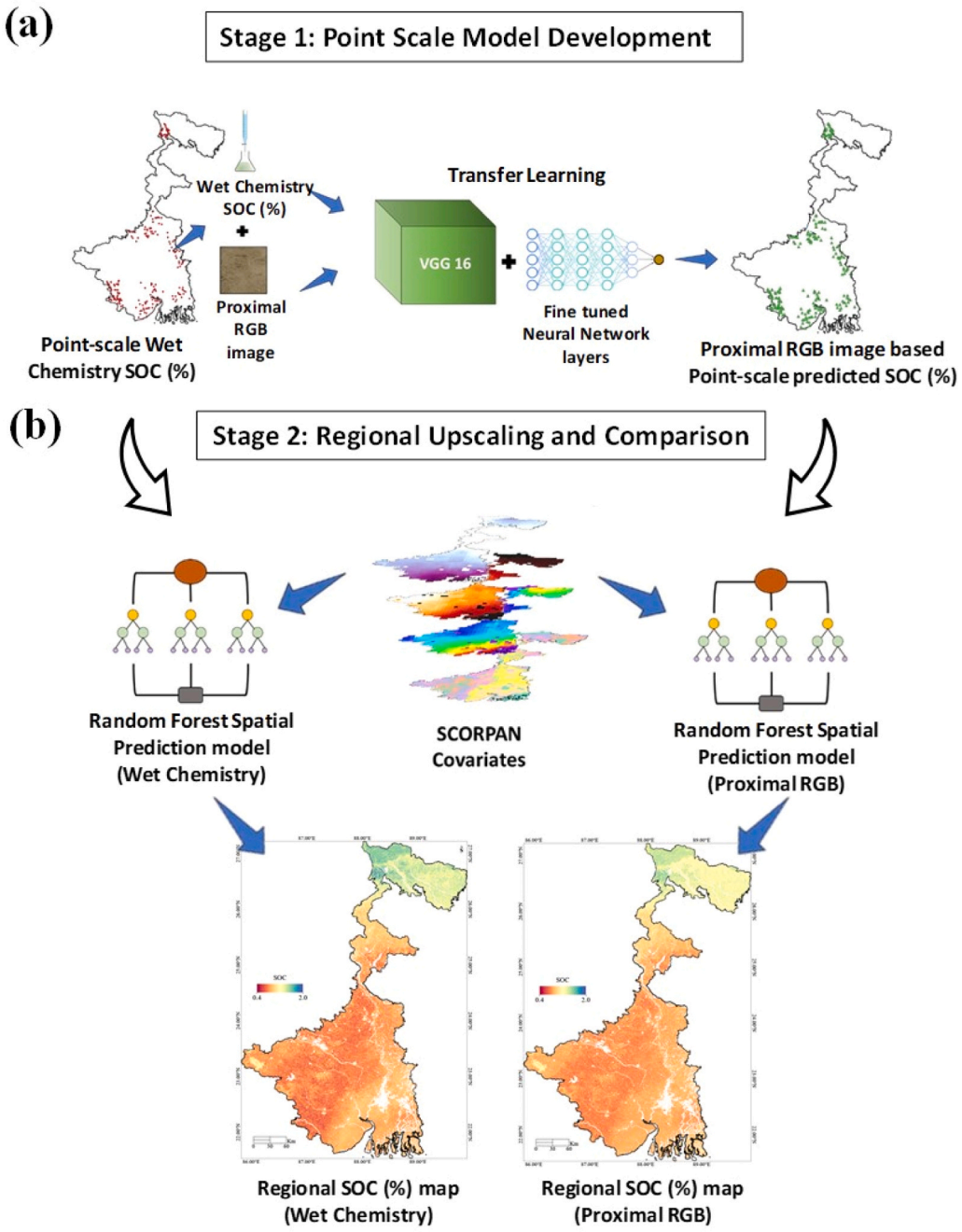


Fig. 3. The two-stage SOC (%) prediction workflow showing (a) Stage 1 (point scale predictions): Estimation of point scale soil organic carbon content using proximal RGB image and VGG16 deep learning-based transfer learning model, and (b) Stage 2 (spatial upscaling): regional prediction of SOC content for the entire state of West Bengal using point scale estimates (from stage 1) and SCORPAN covariates within a digital soil mapping (DSM) framework. For comparison with the proposed two-stage approach, a wet chemistry-based DSM was also implemented.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_{O,i} - y_{P,i}\right)^2}{\sum_{i=1}^{n}\left(y_{O,i} - y'_O\right)^2} \quad (1)$$

$$MAE = \frac{\sum_{i=1}^{n}\left|y_{O,i} - y_{P,i}\right|}{n} \quad (2)$$

where, $y_{O,i}$ and $y_{P,i}$ are observed and predicted values of ith sample, y'_O is the average of observed values, n is the number of observations.

Additionally, soil organic carbon content of test datasets were classified into low (<0.5 %), moderate (0.5–0.75 %) and high SOC (>0.75 %) content (Ramamoorthy and Bajaj, 1969). A confusion matrix was generated to find the agreements between actual and predicted SOC content classes. A brief workflow of the overall experimental workflow is shown in Fig. 3.

2.8. Benefit-Cost analysis of the proposed approach

For the Benefit-Cost (BC) analysis of the proposed two-stage method with the baseline wet chemistry-based method, the Analytical Hierarchy Process (AHP) based Multi Criteria Decision-Making (MCDM) approach (Saaty, 2013) was utilized. Both the estimation methods served as two alternatives. For finding the superior alternative, four criteria were selected, such as monetary cost, time requirement, modelling error, and environmental hazard. The details on the monetary costs of various components are mentioned in Table 4. The labor cost required for soil sample collection and processing (grinding and sieving) is common in both approaches and is valued at 6.4 US$ daily, considering the maximum wage (including variable dearness allowance) of semi-skilled/unskilled supervisory labor (Government of India, 2025). An all-inclusive per-sample cost of 2.28 US$ was considered for SOC estimation through a wet chemistry-based method. While for the proximal RGB-based method, a cost of 105.79 US$ for a smartphone and 56.91 US$ for the imaging setup was taken. Considering a 3–5 years of average life-cycle of the smartphone, the cost involvement for the proximal RGB-based SOC estimation method can be considered as a non-recurring cost component. An advantage of the proposed approach is the reduced time requirement for the SOC estimation. Specifically, for point-scale SOC estimation. The total lead time was computed considering a total of 405 soil samples utilized in the current study. For the wet chemistry-based sample analysis, an average processing rate of 20 samples per day was assumed, while for proximal image acquisition, a rate of 80 samples per day was considered (based on an average of 8 working hours per day). The monthly salary of 493.98 US$ (including House Rent Allowance) for one Research Analyst (RA) was considered as per the stipend guidelines available in CSIR-HRDG (2023) for the WC-based approach (soil sample preprocessing and chemical analysis) as well as the proximal RGB-based approach (RGB photography capturing and subsequent modelling). The total expenditure on the RA salary differed between the methods due to the varying time requirements necessary to complete the 405 sample analysis and subsequent DSM. The time requirements for DSM were kept equal after proximal SOC content estimation using either the WC or the proximal RGB-based approach. Also, as the accuracy of both wet chemistry and proximal image-based SOC estimation methods is expected to vary in terms of accuracy, the MAE of both methods was also utilized as a penalty factor. The environmental hazard emanating from the use of chemicals such as chromic acid and sulphuric acid in the wet chemistry approach was used as a fourth criterion in the MCDM. The environmental impact of the hazardous chemicals was computed based on Aquatic Acute Toxicity, AATP, and Human toxicity potential by ingestion, HTPI (Young and Cabezas, 1999; Serna et al., 2016).

Table 4
Details on the cost of various components in both alternatives.

Alternative	Component Details	Cost	Recurring (R)/ Non-recurring (NR)
Wet Chemistry	Labor cost	6.4 US$ daily (for 3 months) for two persons	R
	Sample analysis cost	2.28 US$ per sample, total 405 samples	R
	Analyst cost	493.98 US$ per month (for 0.67 months) for one person	R
Proximal RGB	Labor cost	6.4 US$ daily (for 3 months) for two persons	R
	Smartphone cost	105.79 US$ (fixed cost)	NR
	Image Set up cost	56.91 US$ (fixed cost)	NR
	Analyst cost	493.98 US$ per month (for 0.16 months) for one person	R

$$AATP = \frac{1}{LC_{50}} \quad (3)$$

$$HTPI = \frac{1}{LD_{50}} \quad (4)$$

Where, LC_{50} (mg L^{-1}) and LD_{50} (mg kg^{-1}) are the Lethal Concentration and Lethal Dose of a specific chemical to 50 % of the test population.

$$\text{Hazard Score} = AATP_{\text{chromic acid}} + HTPI_{\text{chromic acid}} + AATP_{\text{sulphuric acid}} + HTPI_{\text{sulphuric acid.}} \quad (5)$$

For the current study, an LD_{50} of 80 mg kg^{-1} (Ayres and Hellier, 2022) and an LC_{50} of 93.09 mg L^{-1} (Ghosh and Saha, 2022) were utilized for Cr^{6+} exposure. While for sulphuric acid, a value of 0.375 mg L^{-1} and 20 mg kg^{-1} was considered (Takeuchi et al., 2021) as LC_{50} and LD_{50}, respectively. A sum of AATP and HTPI for all components has been used as the fourth criterion.

3. Results

3.1. SOC model prediction performance

The proximal RGB image-based SOC prediction models demonstrated high training accuracy. Among them, the proximal model trained using a fine-tuned VGG16 architecture (PM_{VGG}) achieved 4.3 % higher training R^2 and 18.1 % lower training MAE compared with the model based on average image color features (PM_{COL}) (Fig. 4). Both models exhibited notable reductions in accuracy when evaluated on the validation set (n = 91). Nevertheless, the fine-tuned PM_{VGG} model outperformed PM_{COL} in the validation data, with a 37.5 % higher test R^2 and 13.1 % lower test MAE. Scatterplots for PM_{COL} indicated that the color feature-based predictions rarely exceeded 2 % SOC, particularly in the validation set. The kernel density estimates (KDE) further revealed a higher frequency of underestimation for SOC values < 1 %. In contrast, PM_{VGG} performed better in predicting higher SOC values, as evident from the training scatterplots. This model effectively captured SOC variability at carbon contents exceeding 2 %, particularly in the validation dataset.

Proximal RGB image-based SOC predictions at the point scale were further incorporated within a DSM framework for large scale SOC mapping. For comparing performance of the two-stage approach, another DSM model was also trained using the wet chemistry based SOC content of sampled locations. Notably, while training the DSM models, both the train and validation datasets utilized during proximal model development were considered. A separate test dataset not utilized elsewhere in the modelling before, was used to compare DSM model performances. The performances of the DSM models trained on wet chemistry-based SOC values (DSM_{WC}) and proximal RGB + VGG16

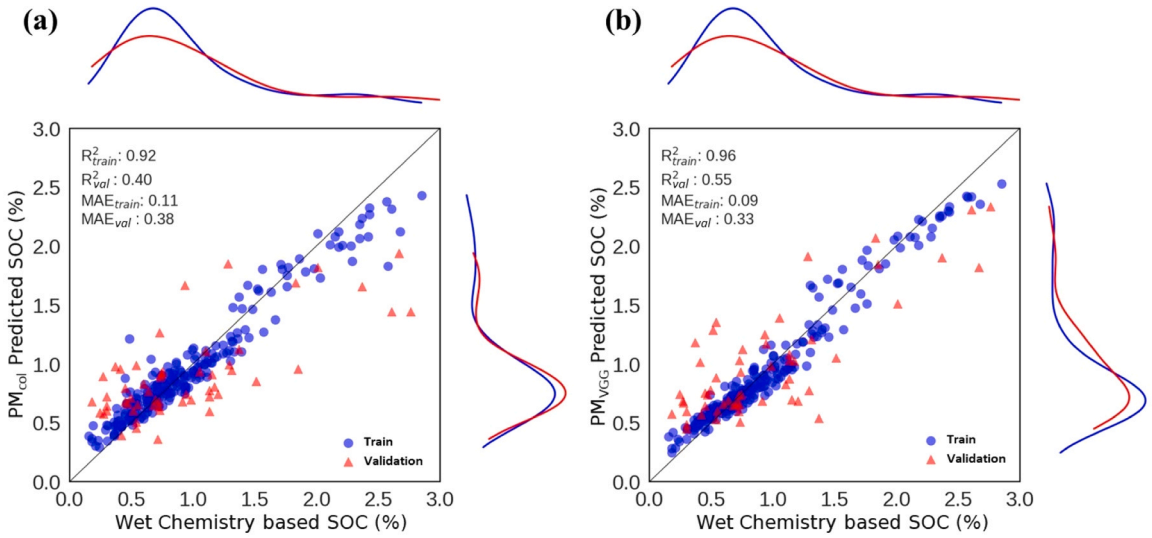


Fig. 4. Scatterplots between observed and predicted SOC (%) for training and validation datasets in (a) Average color feature-based model (PM_{COL}), and (b) VGG16-based model (PM_{VGG}) for soils of West Bengal, India.

predicted SOC values (DSM_{RGB}) (n = 305) are presented in Fig. 5. Both the DSM models, differing only in data modality, showed almost similar accuracy for the training dataset. However, in the test set (n = 100), the DSM_{RGB} model achieved 8.8 % higher R^2 and 32.1 % lower MAE compared with the DSM_{WC} model. The DSM_{WC} model is a single step approach and the error consists of only the DSM based on the wet chemistry data. In contrast, the total error in proximal RGB based DSM model consists of errors from both the PM_{VGG} and DSM_{RGB} models. Although the DSM_{RGB} model produced lower error than DSM_{WC} in the second stage, it also predicted 0.4 % higher SOC content in the lower range of actual SOC and 0.5 % higher in the higher range, relative to DSM_{WC}.

3.2. Important covariates for DSM of SOC

The importance of the five most influential features, based on Shapley values for both DSM_{WC} and DSM_{RGB}, was illustrated through the beeswarm plots in Fig. 6. In both the models, the first four most important features were similar and related to climate, soil, and vegetation properties of the region. In the wet chemistry-based DSM, topographical roughness additionally showed a significant impact on SOC prediction, whereas in the RGB image-based DSM, bare soil indices linked to crop residue cover were more important.

Among climatic variables, mean annual temperature (BIO 01) and annual precipitation (BIO 12) had the most significant influence on SOC predictions. In both the DSM models, higher BIO 01 values were associated with lower SOC predictions. Mean annual temperature near surface is correlated with soil temperature (Amato and Giménez, 2024). As soil temperature increases, soil organic matter decomposition rates also increase. However, the temperature sensitivity is greater at low temperatures than at moderate to high temperatures of $> 20°C$ (Kirschbaum, 1995). In our current study, the contribution of high BIO 01 values to low SOC predictions was modest, with the highest Shapley frequency near 0.1. However, for a few observations, lower BIO 01 values were strongly associated with higher SOC predictions, with Shapley values ranging from 0.55 to 0.62. The effect of BIO 12 was more

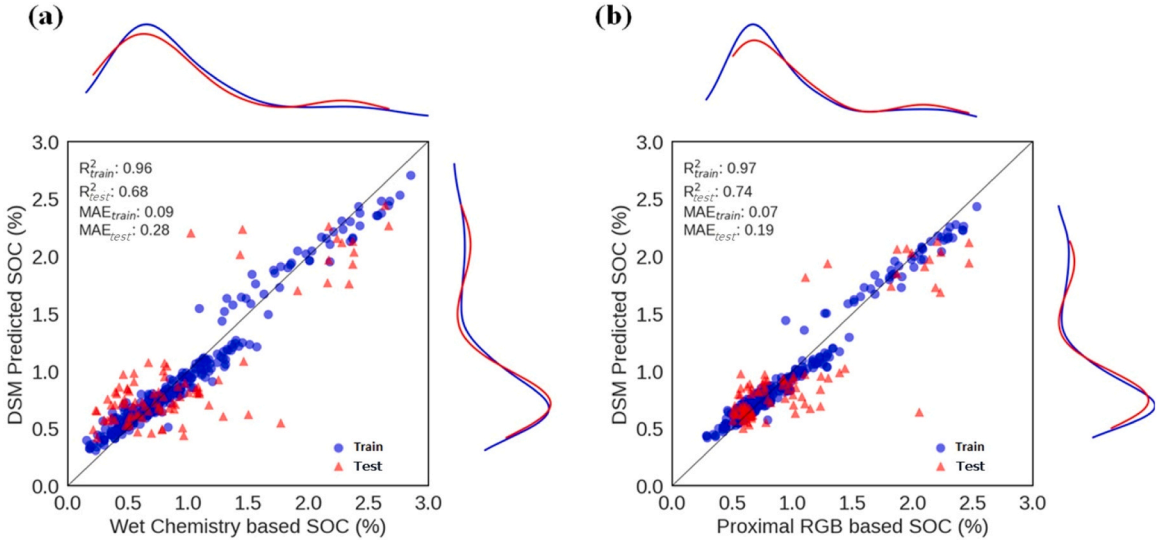


Fig. 5. Scatterplots between observed and predicted SOC (%) for training (n = 305; consisting of both train and validation datasets) and independent test datasets (n = 100) in (a) Wet chemistry-based digital soil mapping model (DSM_{WC}), and (b) Proximal RGB image-based (through VGG16) digital soil mapping model (DSM_{RGB}) using random forest (RF) for soils of West Bengal, India.

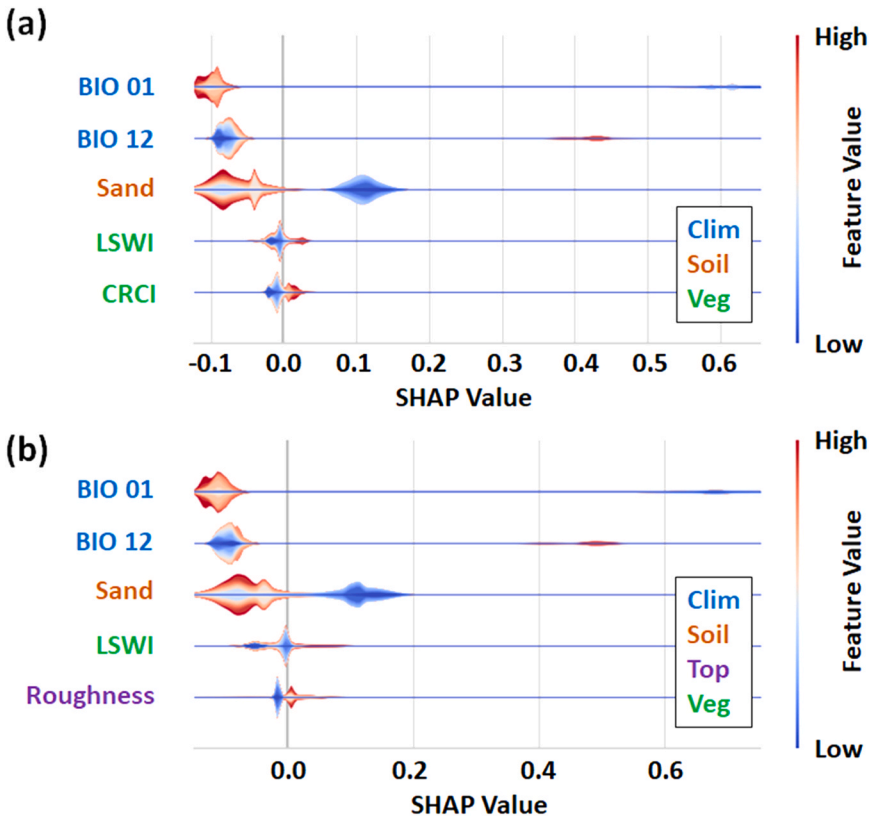


Fig. 6. Shapley contribution of the five most important features in (a) DSM_{RGB} and (b) DSM_{WC} models for soils of West Bengal, India. Clim: Climate, Top: Topography, Veg: Vegetation and Soil: Soil related dominant soil forming factors.

complex. For low SOC observations, predictions were primarily linked to low precipitation rates, although a few cases also corresponded to high BIO 12 values. This suggested that precipitation was not the sole governing factor for low SOC content in certain regions. As the Shapley summary plots reflected individual feature contributions, they did not account for feature interactions.

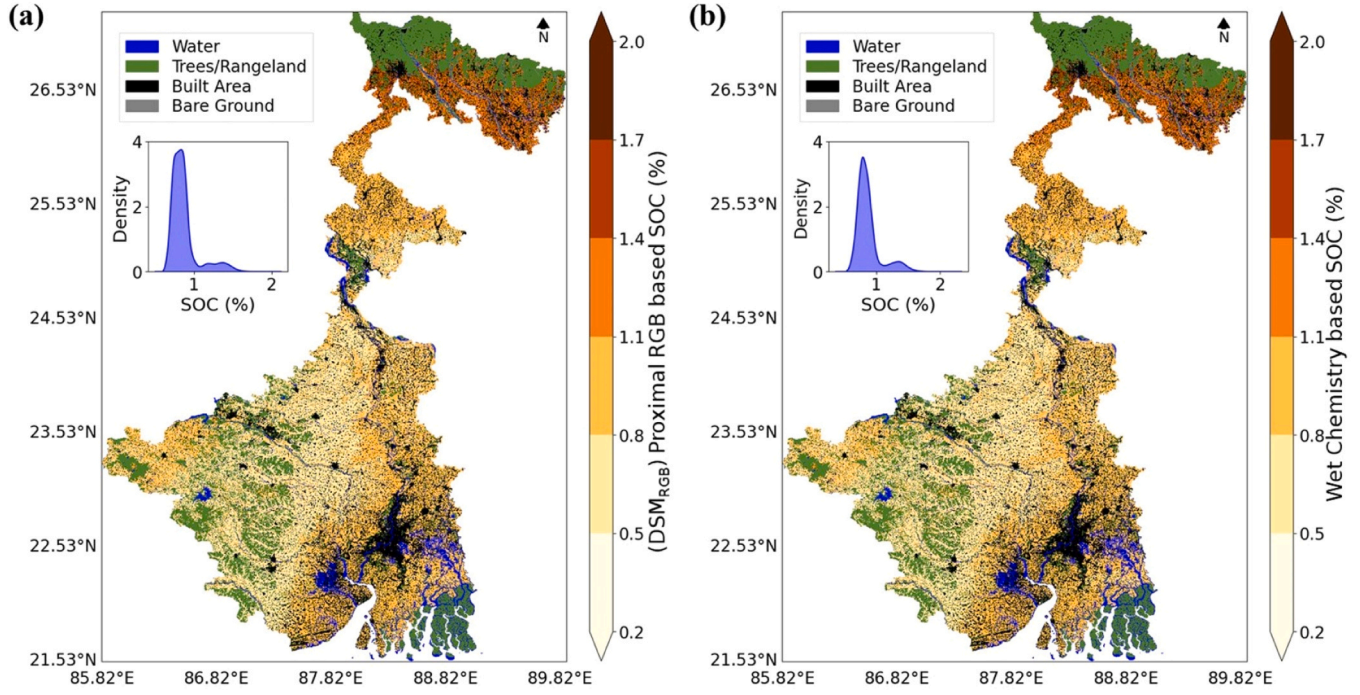


Fig. 7. Spatial distribution of SOC (%) in the agricultural lands of West Bengal, India for (a) Proximal red green and blue (RGB) image- based digital soil mapping (DSM) and (b) wet chemistry (WC) based DSM. Histograms of predicted SOC (%) are shown in the inset. The other dominant land use land cover classes (from ESRI LULC cover dataset, Karra et al., 2021) are also shown.

Among soil properties, sand content was the most influential factor for SOC variation, with higher sand content consistently linked to lower SOC predictions. Areas with lower sand content typically exhibited higher SOC predictions, although in a few cases, low sand content was associated with reduced SOC, indicating the dominance of other pedogenic factors.

Vegetation indices related to crop growth patterns or long-term bare soil reflectance also impacted SOC predictions in both the directions. The low long term (18 years) mean LSWI, derived from Active Crop composite, related to the canopy moisture content and vigor, was generally linked to lower SOC content and vice versa. Soils with low SOC content produces lower water holding capacity (Rawls et al., 2003) and overall fertility (Gerke, 2022), resulting into poorer vegetation growth and may also show low canopy moisture levels in the overlying vegetation (Xing et al., 2025). Similarly, low crop residue cover (CRC) values, indicative of greater soil disturbance from tillage or other agronomic practices, were associated with lower SOC content. The role of surface roughness in the DSM_{WC} model was complex, showing both positive and negative effects on SOC predictions across the region, likely due to interactions with other soil-forming factors.

3.3. Spatial distribution of SOC

Spatial maps of predicted SOC across the entire study area, generated using the DSM methods, were presented in Fig. 7. The spatial variation of SOC content in agricultural landscapes predicted by both DSM_{WC} and DSM_{RGB} followed almost identical patterns. The KDE histograms in both cases exhibited right-skewed distributions, with a higher frequency of values in the lower SOC range (0.8 % in DSM_{RGB} and 0.78 % in DSM_{WC}). Both density estimates also displayed secondary peaks at higher SOC contents, specifically at 1.2 % and 1.4 % for DSM_{RGB}, and at 1.34 % for DSM_{WC}.

Areas with SOC content < 0.8 % corresponded to the Red and Laterite zones (RLZ) and Old Alluvial Zone (OAZ), covering much of the study region. Other agro-climatic zones, such as New Alluvial Zone (NAZ) or Coastal and Saline Zone (CSZ), generally showed SOC values between 0.8 % and 1.1 %. In the Terai Zone (TZ), SOC content was higher, ranging from 1.1 % to 2 %, with values increasing from south to north. In this zone, pixels in the northern part exhibited the highest SOC predictions, primarily due to lower temperatures and higher rainfall, as reflected in the higher Shapley contributions of BIO 01 and BIO 12 in a few cases (SHAP values between 0.4 and 0.6) shown on the right side of Fig. 6a and b. Previous studies have also attributed higher SOC in TZ to environmental factors (Das et al., 2021) and manure applications (Debnath and Ghosh, 2011). In contrast, soils of the RLZ were erosion-prone, variable in texture, and poor in organic matter and other nutrients (Ghosh, 2019; Kundu, 2020).

The spatial differences in predicted SOC content (%) between DSM_{WC} and DSM_{RGB} were shown in Fig. 8. Areas with low to moderate SOC content, particularly in the RLZ, parts of the NAZ, and the CSZ, showed lower prediction values in DSM_{RGB} compared with DSM_{WC}. In contrast, in the TZ and parts of the RLZ, DSM_{RGB} predictions were higher than those from DSM_{WC}. Overall, the spatial maps produced by DSM models trained on wet chemistry-based SOC and proximal RGB-based SOC appeared almost identical. The histogram of prediction differences between the two methods exhibited a symmetric Gaussian distribution, with most values clustered near 0. The range of prediction differences was very small, with the 5th percentile at 0.07 % and the 95th percentile at 0.08 %. These results highlighted the potential of proximal RGB imagery for SOC prediction at the point scale and subsequent upscaling to the spatial scale, achieving performance similar to wet chemistry-based methods.

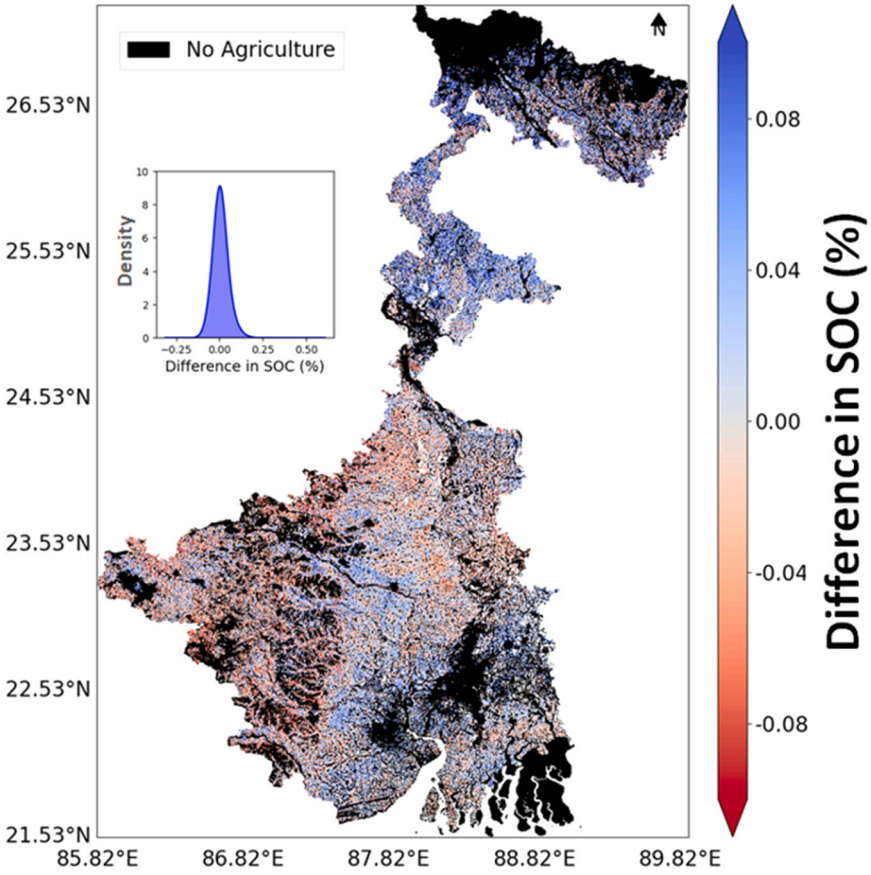


Fig. 8. Spatial map showing the difference in predicted SOC (%) using wet-chemistry (DSM_{WC}) and red, green, and blue (DSM_{RGB}) image-based digital soil mapping in West Bengal, India. Warmer colors represent overestimation of DSM_{RGB}, while cooler tones correspond to underestimation with respect to DSM_{WC}.

SOC predictions from both DSM methods over the test datasets were classified into three categories: low, moderate, and high SOC content. The confusion matrix heatmaps (Fig. 9) indicated very similar performances for both methods. However, observations with low SOC content were frequently misclassified. In DSM_{RGB}, no instances with actual SOC content in the low category were correctly classified, whereas in DSM_{WC}, only 9 % of the actual low SOC observations were correctly classified. This suggested that both DSM methods were unable to effectively model low SOC content in agricultural soils. For the moderate and high SOC categories, both methods showed comparable performance. In DSM_{RGB}, 62 % of samples were correctly classified into the moderate SOC category, and 76 % were correctly classified into the high SOC category. In DSM_{WC}, 66 % of samples were correctly classified into the moderate SOC category, and 69 % were correctly classified into the high SOC category.

Satellite data-based DSM approach offers an additional advantage for predicting temporal changes of SOC content over a certain region. The SOC content spatial distribution maps in the southern-most portion of the study area during 2015–2020 and 2020–2025 are shown in Fig. 10 a and b. The eastern part of the region belonging to partly CSZ and OAZ shows higher SOC content (>0.75 %) in both the time frames. While the western part falling in the RLZ, shows lesser SOC content predominantly within 0.5–0.75 %. The changes in SOC content (δSOC) between 2015 and 20 and 2020–25 are shown in Fig. 10c. The warmer regions correspond to decline in SOC content (up to 0.1 % decline) and are concentrated primarily in the RLZ and also in smaller patches of CSZ and OAZ. While a dominant patch also shows around 0.1–0.15 % increase in SOC, specifically in the OAZ and CSZ and in the eastern part of the RLZ.

The ground samples collected for this subset of study region during both time frames also indicated the predominance of SOC in the medium to high range class (>0.5 % SOC). The two-stage DSM predicted SOC content over this region showed higher accuracy in high SOC class compared to the medium SOC class (Fig. 11). For the 2020–25 timeframe, samples in the medium SOC category showed 15.4 % higher accuracy than in 2015–20. While, in the high SOC category, an 18.8 % decline in correctly classified samples was found in the 2020–25 timeframe from 2015 to 20.

3.4. Benefit-Cost analysis of SOC estimation methods

The cost-effectiveness of two alternative frameworks was evaluated on the basis of four criteria on cost, time requirement, accuracy, and environmental hazards. In Table 5, the BC analysis of the two alternatives for SOC estimates is shown. For the current study, the RGB image-based method was found to be 5.18 times cheaper in cost and 4 times faster than the WC WC-based method. Regarding model accuracy criteria, although the error in the RGB RGB-based method is twofold, the overall error is comparable with the single-stage WC-based method. The WC-based method showed a composite hazard score of 2.75 due to the requirement of chromic acid and sulphuric acid, while the RGB image-based method poses no such environmental threats. The model based on the four criteria was found to be consistent, as evident from the low consistency ratio (CR) of less than 0.1 (CR: 0.02).

Initial pairwise comparison between four criteria produced the highest priority score of 0.51 for model accuracy, while the lowest score of 0.12 was given to environmental hazards. Both cost and time showed a moderate priority score of 0.17 each. During the comparison of two alternatives, WC and proximal RGB-based SOC estimation, a technique with lower cost, time, error, and hazard is considered to have more benefit than the other. The weighted scores of alternatives and the AHP score are shown in Table 6. Across all criteria, the proximal RGB image-based technique showed higher benefit than the WC-based approach in terms of cost and time (4 times lower score for time and 5 times lower score for monetary cost). Although the highest priority was set to accuracy, due to almost comparable MAE, similar scores were set for both alternatives. In the case of the hazard score, the lowest score was set for the WC WC-based method due to its reliance on toxic chemical reagents. The AHP-based MCDA tool identified the proximal RGB-based method as a superior alternative as compared to the WC-based method, due to a lower cost, time, and environmental hazard. The final AHP score showed a significantly higher value of the proximal RGB-based SOC estimation method than the WC-based traditional method.

4. Discussion

4.1. Characterizing distribution of errors in two-stage SOC prediction

Our two-stage SOC prediction framework demonstrated strong overall performance with the deep learning-based model (PM_{VGG}) outperforming the average color feature-based model (PM_{col}) (validation R^2 of 0.55 in PM_{VGG} and 0.4 in PM_{col}). When the PM_{VGG} based SOC predictions are incorporated in the DSM framework, large-area mapping accuracy was found to be comparable to wet chemistry benchmarks. This highlights the value of proximal DL in optimizing traditional methods of SOC monitoring. At the same time, it is essential to evaluate how modelling errors propagate from the proximal modelling stage

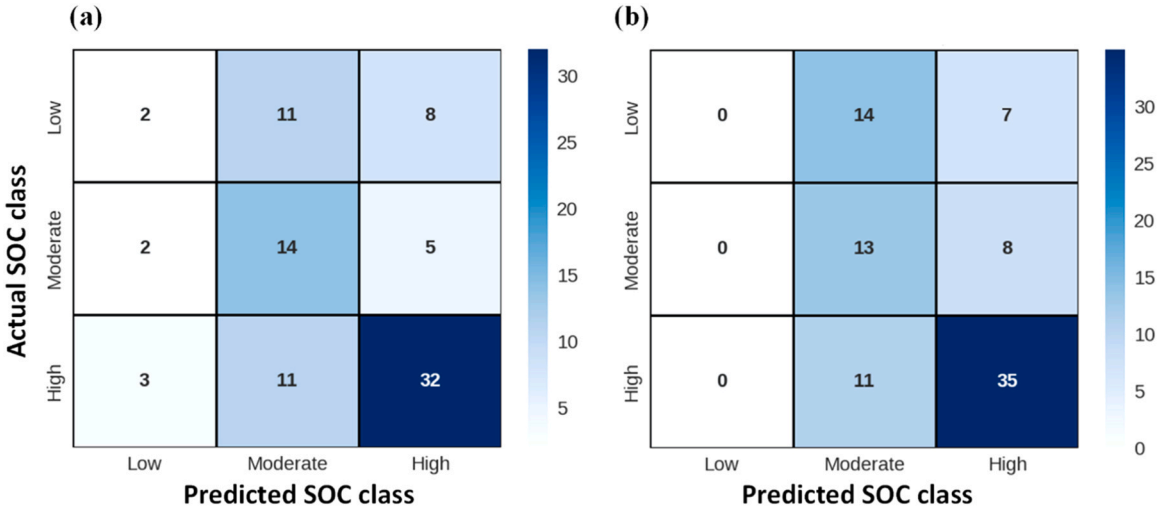


Fig. 9. Confusion Matrix heatmap of (a) wet-chemistry-based digital soil mapping (DSM_{WC}) and (b) Proximal red, green, and blue (RGB) image-based digital soil mapping model (DSM_{RGB}) for soils of West Bengal, India. SOC content (%) was classified into low (<0.5 %), moderate (0.5–0.75 %), and high (>0.75 %) value categories (Ramamoorthy and Bajaj, 1969).

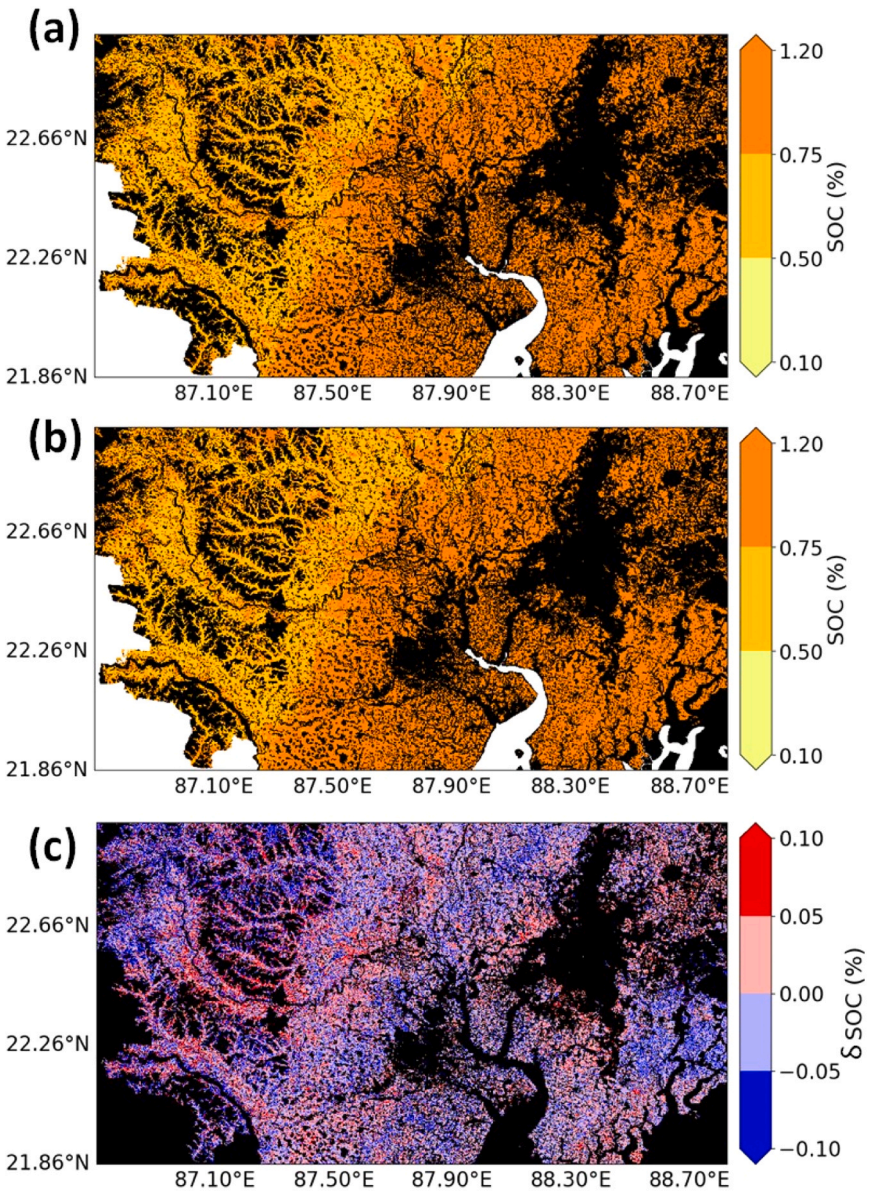


Fig. 10. Predicted SOC content (%) for (a) 2015–2020, and (b) 2020–2025. (c) The difference in SOC content (δSOC in %) between 2015 and 20 and 2020–25.

through the DSM stage to influence the accuracy of final large-area SOC predictions.

Fig. 12a shows the scatterplot between the predicted SOC from the two-stage $PM_{VGG} + DSM_{RGB}$ model and actual wet chemistry-based SOC for both training and testing datasets. This illustrates the total error emanating from proximal RGB image based model for point scale predictions followed by DSM model for large scale SOC prediction, as compared to the wet chemistry based SOC. The proximal RGB-based DSM demonstrated nearly comparable performance (R^2 of 0.7 and MAE of 0.27 %), to the wet chemistry-based DSM (R^2 of 0.68 and MAE of 0.28 %) in the test dataset (n = 100) (Figs. 5a and 12a). The relationship between the residuals of proximal RGB image-based model and final two-stage spatially upscaled model are shown in Fig. 12b. The residuals are calculated as a difference between wet chemistry measured and model predicted SOC (%). The scatterplot shows a strong positive correlation (Pearson's Correlation Coefficient (r): 0.83) between only proximal and two-stage model residuals. To verify the authenticity and robustness of the proposed two-stage SOC prediction framework, multiple independent validation strategies are employed. First, model performance is evaluated using a strictly held-out test dataset (n = 100) that is not used during proximal model training or DSM calibration, ensuring an unbiased assessment of predictive capability. Second, proximal RGB image-based SOC predictions are benchmarked against laboratory wet chemistry measurements using standard accuracy metrics (R^2 and MAE), and their propagation through the DSM framework is explicitly analysed via residual decomposition. The strong linear relationship (r = 0.83) observed between proximal prediction residuals and final DSM residuals indicates that spatial upscaling preserves the underlying error structure rather than artificially inflating model performance. Third, the spatial SOC maps generated using RGB-derived predictions are compared with those obtained from wet chemistry-based DSM, revealing near-identical spatial patterns and a narrow distribution of prediction differences (5th–95th percentile range of 0.07–0.08 % SOC). In addition, confusion matrix analysis demonstrates consistent class-level behaviour across both mapping approaches. Finally, temporal validation using independent soil observations collected during two separate time windows (2015–2020 and 2020–2025) further confirms the stability of the trained DSM model. Together, these complementary validation exercises provide strong evidence for the authenticity, reproducibility, and operational reliability of the proposed proximal

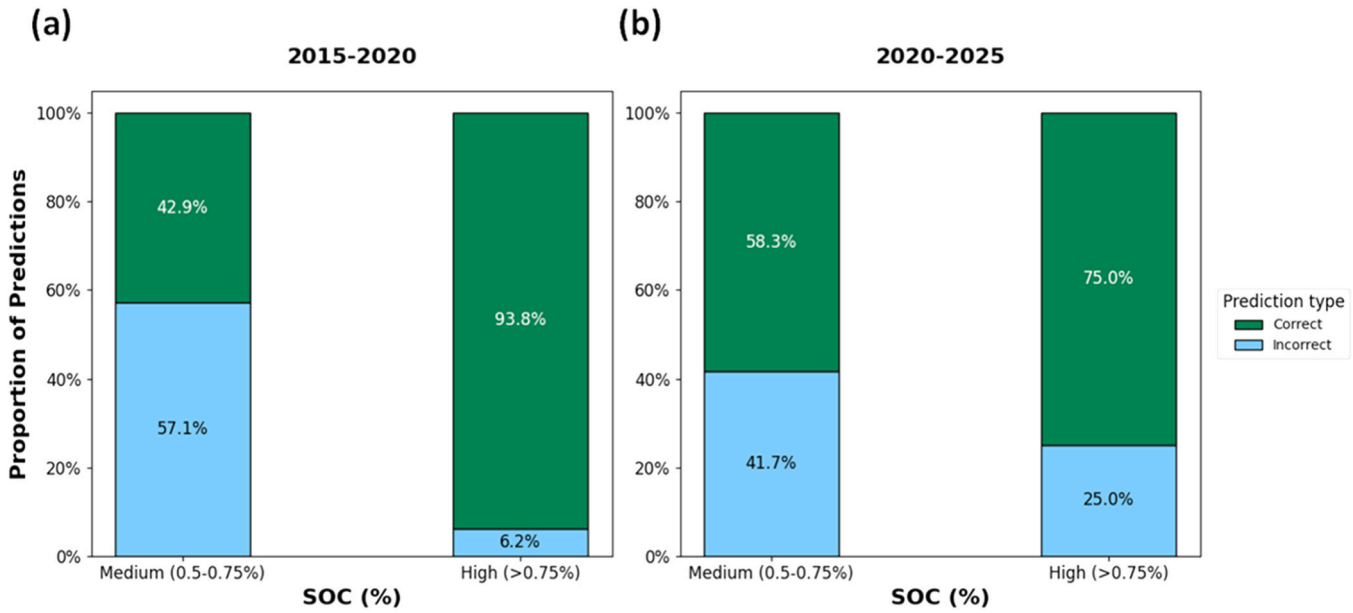


Fig. 11. Proportions of samples showing correct and incorrect prediction in medium (0.5–0.75 % SOC) and high (>0.75 % SOC) class for (a) 2015–2020 and (b) 2020–2025 time frame. The sample size was 27 in 2015–20 and 36 in 2020–25.

Table 5
BC analysis of WC and proximal RGB image-based SOC estimation methods.

Criterion	WC value	RGB value
Cost	1254.37 US$ (3.09 US$ per sample)	241.74 US$ (0.59 US$ per sample)
Time	20.25 days	5.06 days
Error	0.28 %	0.27 %
Hazard	2.75	0.001

Table 6
Weighted score and AHP score analysis result for two different alternatives.

Alternative	Error	Cost	Time	Hazard	AHP score
Wet Chemistry	0.25	0.03	0.04	0.0001	0.32
Proximal RGB	0.26	0.16	0.16	0.07	0.67

RGB–DSM framework.

A slope of 0.8 in the fitted linear regression line suggests a slight attenuation of PMVGG based point-scale prediction error during spatial upscaling. This likely results from the filtering of random, localized variations inherent in point-scale proximal SOC predictions. When mapping at 100 m spatial grids using satellite covariates, this process leads to spatial smoothing rather than an actual reduction in modelling error. Both the training and testing datasets show similar relationships between residuals of proximal model and final DSM model. Also, a symmetric distribution of points across positive and negative residuals suggests no systematic bias in the model predictions. The higher residuals at low SOC values (<1 % SOC), suggests that the model may underperform for soils with minimal organic carbon content—possibly due to limitations in RGB imaging sensitivity for low-carbon samples.

The distribution of residuals for PM_{VGG}, DSM_{RGB}, and total residual is also shown in Fig. 13 for low (<0.5 %), moderate (0.5–0.75 %), and high (>0.75 %) SOC content ranges. The frequency distribution indicated that the major contribution to the total model uncertainty was driven by SOC predictions from RGB images via the VGG16 model rather than the DSM model. Specifically, the model

overestimated SOC content in both the low and moderate SOC ranges, with median residual errors ranging from 0.15 % to 0.25 %. In contrast, in the high SOC range, the VGG16 model underestimated the wet chemistry-based SOC content, with residuals centered around 0.2 % SOC.

4.2. Robustness of the proposed approach: impact of surface interference

Soil color is linked to soil organic matter content and many researchers used either Munsell colorimetric systems (Company, 1994) or soil color parameters for SOM quantification from digital images (Rossel et al., 2008; Aitkenhead et al., 2015; Sudarsan et al., 2016). Application of deep learning methods can extract features hierarchically and aid in complex decision making (LeCun et al., 2015). The convolutional layers can extract information on texture, shape, patterns and colors (Gyasi and Purushotham, 2023) weights and are utilized mainly on soil spectroscopy data for inferring various soil properties (Zhong et al., 2021; Kasaragod et al., 2024). Gao et al. (2024) utilized the pre-trained MobileNet V2 model and fine-tuned it for soil organic matter prediction. However, the accuracy was found to be less as compared to ML models such as Random Forest Classification, and Convolutional Neural Network model due to a smaller number of training data availability.

While deep learning models show promise in controlled settings, their application to field-collected RGB images is often limited by extraneous factors such as soil wetness condition, presence of surface residues, roughness and lightness conditions (Ben-Dor et al., 2008) may impact the RGB image-based SOM prediction model accuracy. Among these factors, soil wetness is perhaps the most important factor that hinders the practical use of soil color for soil organic matter under field conditions (Stevens et al., 2008; Nocita et al., 2013; Fu et al., 2020). Especially, there is a critical soil moisture content, typically in the order of 15–20 % (Lobell and Asner, 2002; Nocita et al., 2013) above which soil reflectance pattern changes in response to soil organic matter change than what is below that threshold. Also, the decrease in soil spectral reflectance with increasing soil wetness is non-linear in nature (Lee et al., 2003). Nevertheless, in our current study, we have utilized processed and air-dried soil samples to reduce the impact of soil moisture on soil color variations, which may complicate subsequent SOC estimation. Apart from moisture, slight changes in lighting conditions can cause drastic change in soil color parameters for the same soil samples and can hamper the model accuracy (Fu et al., 2020). In field condition, uncontrolled lighting due to changes in sunlight angle, or cloud cover variation can cause shadows, hotspots, and color shifts,

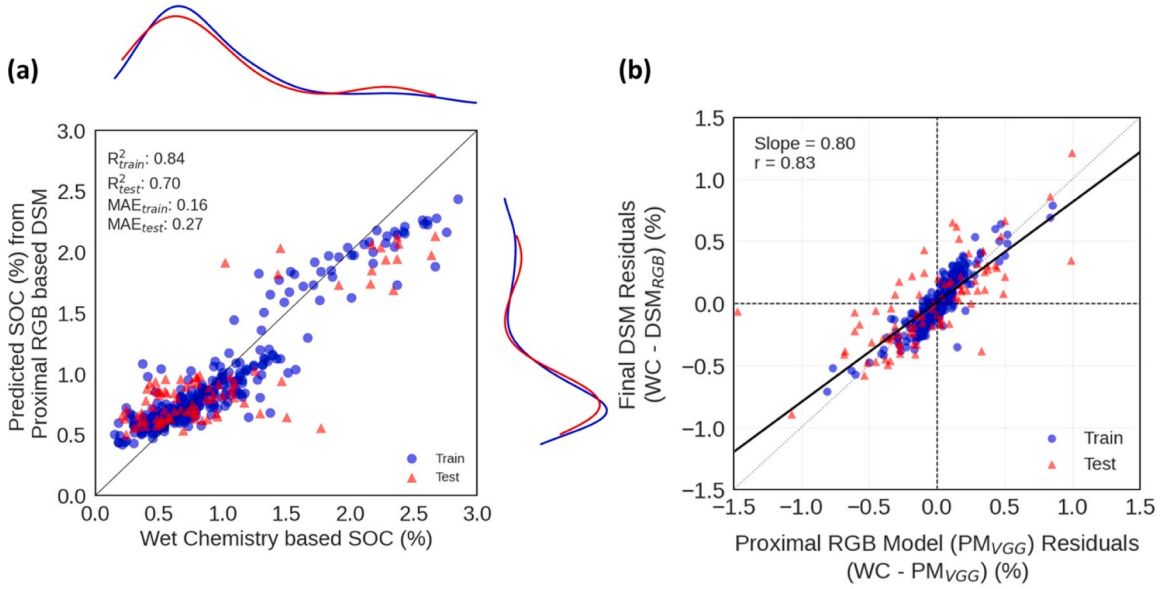


Fig. 12. (a) Scatterplot between wet chemistry-based SOC (%) and predicted SOC (%) through two-step modelling (PM_{VGG} followed by DSM_{RGB}) for soils of West Bengal, India. (b) Relationship between final DSM residuals (%) of two-stage predictions (wet chemistry or WC measured SOC – DSM_{RGB} model predicted SOC) and proximal RGB based SOC residuals (%), calculated as (WC measured SOC – PM_{VGG} model predicted SOC).

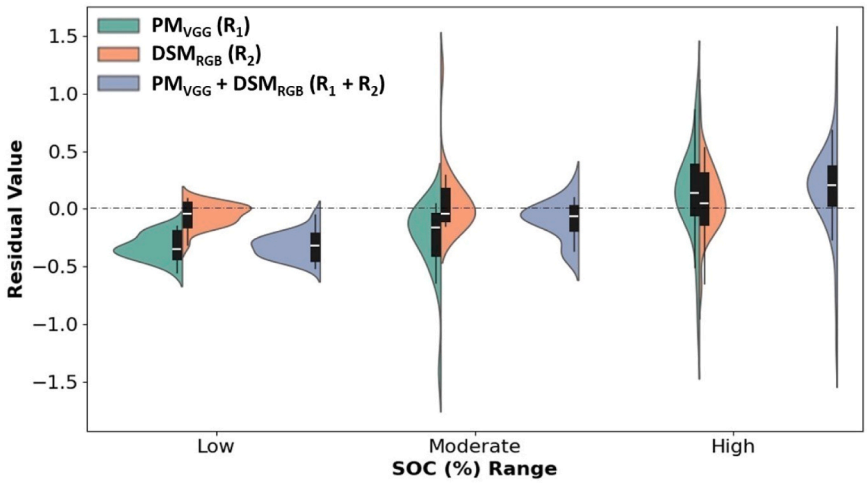


Fig. 13. Frequency distribution histograms and box-whisker plots of residual values for models, PM_{VGG} (R_1), DSM_{RGB} (R_2) and total residual (PM_{VGG}+DSM_{RGB}: R_1+R_2) are shown for low, moderate and high SOC (%) ranges. The black colored box overlaid on each histogram shows the inter-quartile range (IQR), and the white horizontal line inside the box corresponds to the median residual value. The vertical black whiskers extend up to 1.5 times the IQR and data beyond these whiskers represent outliers.

exacerbating inconsistencies in RGB capture and may require additional normalization or perturbation (Gorthi et al., 2021) to enhance robustness. Although images were acquired under controlled laboratory conditions in our current method (with white reference cards and fixed light intensity), model accuracy is likely to suffer if applied to RGB images captured under varying light intensities. Presence of surface residue or debris while capturing soil image in the field may introduce non-soil pixels within the image footprint and introduce additional noise, which needs to be removed beforehand by applying semantic segmentation and exclusion of non-soil pixels (Naeimi et al., 2024). Soil micro-topographical variations can change the roughness of soil and impact the soil spectral reflectance even under laboratory conditions (Herodowicz, 2016). Soils with higher amounts of clay appear to be darker in color as compared to a sandy soil, even when passed through the same sieve size (Ben-Dor et al., 1999). Furthermore, a proximal RGB image-based SOC prediction model developed using a specific smartphone camera may not be directly applicable to other devices. Significant differences in sensors' spectral response and image signal processing algorithms across smartphones can substantially impact image quality (Hong et al., 2001; Yang et al., 2021). Thus, a model built on one device cannot be utilized for SOC prediction using images from another without a color normalization scheme. As smartphone camera technology continues to advance, these discrepancies may widen, further complicating the universal application of smartphone based SOC prediction models (Yang et al., 2021).

4.3. *Practical implications and limitations*

The practical implications of this research were substantial, as the study demonstrated that proximal RGB soil imaging combined with deep learning could serve as a low-cost, scalable alternative that achieves comparable SOC mapping accuracy to wet chemistry-based DSM,

without compromising much on mapping accuracy and enabling rapid in-situ estimates in resource-limited settings. By using RGB imagery from consumer-grade mobile devices, the approach enabled rapid, SOC estimation at the point scale, which could be effectively upscaled into spatial digital soil maps using environmental covariates within the DSM framework. The near-equivalence in performance between RGB-based DSM and wet chemistry-based DSM, with differences in predicted SOC values typically within 0.07–0.08 %, suggested strong potential for operational deployment in regions lacking laboratory infrastructure. This method could support precision agriculture by enabling site-specific SOC management, informed farm carbon accounting for climate change mitigation, and facilitate large-scale soil health monitoring programs (Viscarra Rossel et al., 2010; Vågen and Winowiecki, 2013). Furthermore, the portability and low-cost of the image acquisition setup could make SOC mapping more accessible in resource-constrained settings, aligning with global efforts toward sustainable soil management and meeting monitoring requirements under international climate commitments.

Nonetheless, certain limitations were evident. An inherent limitation of this study arose from the spatiotemporal scale inconsistency between point-scale SOC measurements and proximal RGB image predictions at centimetre resolution, and the gridded environmental covariates used for digital soil mapping, which ranged from 30 m to 1 km in spatial resolution and represented multi-year temporal averages. Such scale mismatches were well recognized in digital soil mapping, as soil observations captured local pedological conditions while satellite-derived covariates represented aggregated landscape-scale processes. In the proposed two-stage framework, this inconsistency was mitigated by adopting a hierarchical modelling strategy in which fine-scale SOC variability captured through proximal sensing was first preserved at the point level and subsequently embedded within broader soil-forming processes represented by SCORPAN covariates during spatial upscaling. The use of long-term climatological means and multi-year vegetation composites allowed the models to represent quasi-stationary environmental controls on SOC formation and stabilization, thereby reducing temporal noise associated with short-term weather variability. Furthermore, all covariates were harmonized to a common 100 m grid prior to modelling, ensuring spatial consistency across datasets. Nevertheless, residual scale effects may still have contributed to prediction uncertainty, particularly in heterogeneous landscapes and for low-SOC soils, and future work should explore the integration of finer-resolution climate products, field-scale remote sensing, and spatiotemporal modelling approaches to further reduce scale-induced bias.

Furthermore, both DSM methods performed poorly in accurately predicting low SOC content (<0.5 %), with DSM_{RGB} failing to correctly classify any low SOC samples and DSM_{WC} achieving only 9 % accuracy in this category. Also, under field-condition, the RGB image based –SOC prediction is also sensitive to surface interferences (moisture, lighting, and residue cover), which could further reduce overall prediction accuracy (Fu et al., 2020). Potential solutions include incorporating additional proximal sensing data (e.g., portable spectroscopy, multispectral sensors) to improve low SOC predictions, applying illumination correction and moisture normalization techniques during image acquisition, and retraining models with diverse datasets from multiple agro-climatic zones to enhance robustness. Future work should also focus on validating the approach with unprocessed field samples and integrating it with satellite-based DSM for multi-scale SOC monitoring.

5. Conclusions

This study demonstrates the feasibility of integrating proximal RGB soil imaging with DL and ML-based DSM for large-area estimation of SOC in agricultural landscapes. The proposed two-stage framework shows that consumer-grade smartphone imagery, when combined with a transfer-learning-based VGG16 model and spatial environmental covariates, can serve as a scalable alternative to conventional laboratory-driven SOC mapping workflows, with spatial prediction performance comparable to wet chemistry-based DSM. The modelling results provide new insights into the dominant environmental controls on SOC distribution across heterogeneous agro-climatic zones, where mean annual temperature, precipitation, soil texture, and vegetation water status emerge as key drivers governing regional SOC variability. The benefit–cost analysis further indicates that proximal RGB-based SOC estimation offers a substantially more economical and time-efficient alternative to laboratory-based methods while eliminating the environmental hazards associated with chemical reagents, making it attractive for precision agriculture, soil health surveillance, and regional carbon accounting. Nevertheless, reduced sensitivity for very low SOC soils, sensitivity to surface conditions such as moisture, lighting, residue cover, and device-specific camera characteristics, and residual scale effects arising from the integration of point-scale proximal data with coarser-resolution environmental covariates remain important limitations. Future research should therefore focus on validation using unprocessed field samples, integration of complementary proximal sensors to improve low-SOC sensitivity, and evaluation of model transferability across different pedoclimatic regions and cropping systems. Overall, the study establishes a robust and cost-effective pathway for regional SOC monitoring using low-cost proximal sensing and artificial intelligence, providing a strong foundation for next-generation digital soil monitoring systems in support of sustainable land management and climate-smart agriculture.

CRediT authorship contribution statement

Shubhadip Dasgupta: Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation. **Somsubhra Chakraborty:** Writing – original draft, Supervision, Project administration, Formal analysis, Data curation, Conceptualization. **Bimal K. Bhattacharya:** Writing – original draft, Supervision, Project administration, Conceptualization. **David C Weindorf:** Writing – original draft, Supervision, Project administration, Conceptualization. **Sanjay Srivastava:** Writing – original draft. **Ayan Das:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Atreya Sinha:** Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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